

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1423

Commonwealth Chesapeake 138 kV
September 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1423. The Interconnection Customer proposes a 218.7 MW battery energy storage facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on March 12, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 445 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,768,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W -Station-St Bonifacius 138 KV
Contingency: Southtown-Vienna Operations 345 KV
Loading: 283.7% (Rating: 417 MVA)
DFAX: 53.6%
Pre-project loading: 255.6%
2. Facility: Chemoite-Randall County 138 KV
Contingency: Vienna Operations-Easton 2 345 KV
Loading: 124.8% (Rating: 230 MVA)
DFAX: 15.3%
Pre-project loading: 110.2%
3. Facility: Arroyo Tap Substation-Mesquite 138 KV
Contingency: Southtown-Vienna Operations 345 KV
Loading: 116.5% (Rating: 859 MVA)
DFAX: 46.2%
Pre-project loading: 104.8%
4. Facility: Ocean City Substation-Reidland 138 KV
Contingency: Southtown-Vienna Operations 345 KV
Loading: 116.0% (Rating: 878 MVA)
DFAX: 46.2%
Pre-project loading: 104.5%
5. Facility: Arroyo Tap-Reidland 138 KV
Contingency: Southtown Substation-Vienna Operations 345 KV
Loading: 116.0% (Rating: 878 MVA)
DFAX: 46.2%
Pre-project loading: 104.5%

6. Facility: Candelaria Arenas 1718-Ocean City 138 KV
 Contingency: Southtown-Vienna Operations 345 KV line
 Loading: 115.6% (Rating: 893 MVA)
 DFAX: 46.2%
 Pre-project loading: 104.2%
7. Facility: Carbon Ctr 500-345 kV TX
 Contingency: Shawville Station-Blackrock 345 KV
 Loading: 107.5% (Rating: 1,998 MVA)
 DFAX: 6.2%
 Pre-project loading: 106.8%
8. Facility: Carbon Center-Shaville Sta 500 KV
 Contingency: Blackrock-ShawvilleSta Substation 345 KV line
 Loading: 101.2% (Rating: 2,111 MVA)
 DFAX: 6.2%
 Pre-project loading: 100.6%
9. Facility: Ess Lewes-Newmark 138 KV
 Contingency: Easton 2-Vienna Operations 345 KV
 Loading: 101.0% (Rating: 368 MVA)
 DFAX: 25.1%
 Pre-project loading: 86.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W -Station-St Bonifacius 138 KV	Rebuild line with higher-capacity conductor	\$6,559,000	8% (\$546,000)
Chemolite-Randall County 138 KV	Rebuild line with higher-capacity conductor	\$7,405,000	8% (\$630,000)
Arroyo Tap Substation-Mesquite 138 KV	Rebuild line with higher-capacity conductor	\$2,693,000	9% (\$237,000)
Ocean City-Reidland 138 KV	Rebuild line with higher-capacity conductor	\$5,175,000	9% (\$456,000)
Arroyo Tap-Reidland 138 KV	Rebuild line with higher-capacity conductor	\$937,000	9% (\$83,000)
Candelaria Arenas 1718-Ocean City 138 KV	Rebuild line with higher-capacity conductor	\$3,451,000	9% (\$304,000)
Carbon Center 500-345 kV XFMR	Replace transformer with higher-rated unit	\$33,016,000	4% (\$1,350,000)
Carbon Center-Shaville Sta 500 KV	Reconductor line	\$46,106,000	4% (\$1,886,000)
Ess Lees-Newmark 138 KV L	Reconductor line	\$3,230,000	8% (\$276,000)
	Total allocated to project		\$5,768,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.